Agile Methodologies

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Agile Development: Brief History

- First appeared in 1995.
- The once-common perception that agile methodologies are nothing but controlled code-&-fix approaches, with little or no sign of a clear-cut process, is only true of a small – albeit influential – minority.
- Essentially based on practices of program design, coding and testing that are believed to enhance software development flexibility and productivity.
- Most agile methodologies incorporate explicit processes, although striving to keep them as lightweight as possible.

Major Agile Methodologies

- DSDM – Dynamic Systems Development Method (1994..2014)
- Scrum (1995..2020)
- XP – Extreme Programming (1996, 1999, 2004, 2013)
- ASD – Adaptive Software Development (1997)
- Crystal Family: Orange, Orange Web, Clear (1998, 2001, 2004)
- FDD – Feature-Driven Development (1999, 2002)
- AUP – Agile Unified Process (2005)
- DAD – Disciplined Agile Delivery (2012, 2020)

Agile Methodologies: Agile Manifesto

- The Agile Manifesto is a philosophy — not a strict methodology.
- It encourages **collaboration**, **adaptability**, and **delivering value quickly** to customers while maintaining high-quality software and healthy, motivated teams.

Agile Manifesto

- The Agile Manifesto states that while there is value in the items on the right, Agile values the items on the left more:
- **Individuals and interactions** over **processes and tools**
  → Focus on collaboration, communication, and teamwork rather than relying too much on formal processes or tools.
- **Working software** over **comprehensive documentation**
  → Deliver real, usable software frequently rather than spending too much time on paperwork or documentation that doesn't provide immediate value.
- **Customer collaboration** over **contract negotiation**
  → Engage with customers continuously to refine requirements instead of sticking rigidly to initial agreements.
- **Responding to change** over **following a plan**
  → Be flexible and adapt to new information or shifting requirements, rather than treating a plan as unchangeable.

Agile Methodologies: Principles

- Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.
- Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.
- Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.
- Business people and developers must work together daily throughout the project.

Agile Methodologies: Principles

- Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.
- The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.
- Working software is the primary measure of progress.
- Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Agile Methodologies: Principles

- Continuous attention to technical excellence and good design enhances agility.
- Simplicity—the art of maximizing the amount of work not done—is essential.
- The best architectures, requirements, and designs emerge from self-organizing teams.
- At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behaviour accordingly.

DSDM – Dynamic Systems Development Method

- One of the earliest Agile methods, developed in the UK.
- Focuses on delivering business value quickly and iteratively.
- Uses the **MoSCoW** prioritization technique (Must have, Should have, Could have, Won't have).
- Emphasizes active user involvement and fixed time, variable scope principles.
- Later evolved into **AgilePM**, a project management approach still used today.

Scrum

- Created by Ken Schwaber and Jeff Sutherland.
- One of the most popular Agile frameworks.
- Based on short, time-boxed iterations called Sprints (typically 2–4 weeks).
- Defines clear roles: Product Owner, Scrum Master, and Development Team.
- Focuses on empirical process control (transparency, inspection, and adaptation).

XP – Extreme Programming

- Created by Kent Beck, Ward Cunningham, and Ron Jeffries.
- Focuses on engineering practices for high-quality software and fast feedback.
- Key practices include Test-Driven Development (TDD), Pair Programming, Continuous Integration, and Refactoring.
- Prioritizes customer satisfaction and continuous improvement.

ASD – Adaptive Software Development

- Created by Jim Highsmith.
- Emphasizes adaptation over prediction — fitting Agile's philosophy of responding to change.
- Based on a speculate–collaborate–learn cycle instead of plan–design–build.
- Focuses on complex systems and continuous learning through iteration.

Crystal Family

- Developed by Alistair Cockburn.
- A family of methodologies customized by project size and criticality (e.g., Crystal Clear for small teams).
- Prioritizes people, communication, and reflection over processes or tools.
- Stresses that different projects need different methods — there's no one-size-fits-all approach.

FDD – Feature-Driven Development

- Created by Jeff De Luca and Peter Coad.
- Focuses on designing and building features — small, client-valued functions.
- Uses a five-step process (e.g., develop an overall model, build feature list, plan by feature, design by feature, build by feature).
- Combines model-driven development with iterative delivery.

AUP – Agile Unified Process

- A simplified, Agile version of the Rational Unified Process (RUP) by Scott Ambler.
- Combines Agile practices with the structured discipline of RUP.
- Uses iterative development, test-driven design, and continuous integration.
- Designed to be lightweight, adaptable, and easy to understand.

DAD – Disciplined Agile Delivery

- Also created by Scott Ambler.
- Expands on AUP to cover end-to-end delivery, not just development.
- Integrates ideas from Scrum, Kanban, XP, and Lean.
- Emphasizes context-based choice — helping organizations tailor Agile methods to their needs.
- Became part of the Disciplined Agile Toolkit, now supported by Project Management Institute (PMI).

References

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